

AodeMar: Attention-Aware Occlusion Detection of Vessels for Maritime Autonomous Surface Ships

Ning Wang^{ID}, Senior Member, IEEE, Yuanyuan Wang, Yuan Feng^{ID}, and Yi Wei

Abstract—For maritime autonomous surface ships (MASS), challenges exist in visual detection of occluded marine vessels since invisible occlusion is required to be inferred from locally unoccluded features which are weakly related to the entirety. In this paper, an attention-aware occlusion detection scheme of marine vessels, termed AodeMar, is originated from the viewpoint of MASS transportation. To this end, a position enhancement module is created by virtue of residual connections and coordinate attentions such that high-level semantics and spatial feature dependencies can be efficiently exploited, respectively, thereby accurately locating bounding boxes. Moreover, a multi-scale feature semantics correlation block is devised by combining spatial pyramid pooling and swin transformer-based self-attention encoder in order that the classification ability can be fertilized in both global and local sense. Experiments and comparisons show that the proposed AodeMar outperforms typical approaches including Faster R-CNN, SSD and YOLO series in terms of detection accuracy and robustness.

Index Terms—Occlusion detection of marine vessels, attention-aware occlusion detection, coordinate attention, swin transformer, self-attention encoder.

I. INTRODUCTION

WITH prosperous development of machine vision and deep learning techniques applied to marine vehicles and robotics [1], [2], [3], maritime autonomous surface ships (MASS) are expected to be equipped with visual recognition devices which can facilitate autonomously perceive navigational environment. In this context, informative features of

Manuscript received 17 May 2023; revised 28 October 2023 and 4 April 2024; accepted 6 May 2024. Date of publication 22 May 2024; date of current version 4 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U23A20680 and Grant 52271306, in part by the General Project of National Defense Basic Science Research Program of China under Grant JCKY2022410C013, in part by the National Plan for the Special Support for Top-Notch Talents of China under Grant SQ2022QB00329, in part by Liaoning Leading Talent Project under Grant XLYC2202005, in part by the Major Basic Research Project of Dalian Science and Technology Innovation Fund under Grant 2023JJ11CG009, and in part by the Fundamental Research Funds for Central Universities under Grant 3132023501. The Associate Editor for this article was J. Hemanth. (Corresponding author: Ning Wang.)

Ning Wang, Yuan Feng, and Yi Wei are with the School of Marine Engineering, Dalian Maritime University, Dalian 116026, China, also with the State Key Laboratory of Maritime Technology and Safety, Dalian 116026, China, and also with Dalian Key Laboratory of Green Power Control and Test for Intelligent Ships, Dalian 116026, China (e-mail: n.wang@ieee.org; fengyuan@dlmu.edu.cn; weiyi@dlmu.edu.cn).

Yuanyuan Wang is with the School of Marine Electrical Engineering, Dalian Maritime University, Dalian 116026, China, also with the State Key Laboratory of Maritime Technology and Safety, Dalian 116026, China, and also with Dalian Key Laboratory of Green Power Control and Test for Intelligent Ships, Dalian 116026, China (e-mail: y.wang1010@dlmu.edu.cn).

Digital Object Identifier 10.1109/TITS.2024.3398733

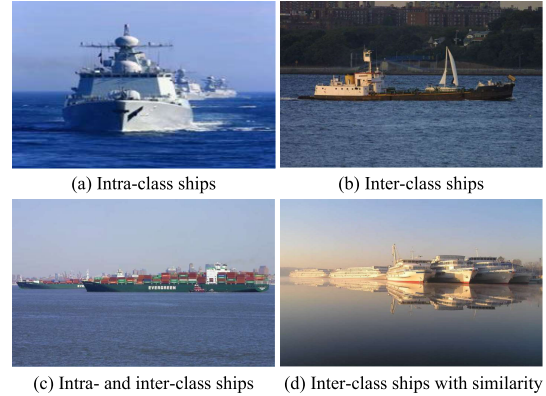


Fig. 1. Overlapping occlusion illustrations of marine vessels.

surface objects for situational awareness derived from deep learning-based image recognition play critical roles in path planning [4], collision avoidance [5], and object tracking [6] of the MASS. To be specific, marine vessel detection takes an essential part in developing MASS intelligent navigation and control systems [7], [8]. Visual detection of marine vessels can be categorized into vertical- and horizontal-view detection [9], [10], [11], [12], whereby the former focuses on externally macro detection and is generally photographed by aircraft unmanned aerial vehicles, while the latter pursues internally micro perception and is captured from the viewpoint of surface vehicles thereby suffering from occluded invisibility of ships. As a consequence, for the MASS, visual-perception autonomy depends in essence on reliable ability of occlusion detection which is desired to be sufficiently robust to marine environments.

Generally, deep learning-based object detectors can be categorized into two- and one-stage approaches. The former generates a series of region proposals [13] which are classified by the convolution neural networks (CNNs). Typical methods include R-CNN series, e.g., Fast R-CNN [14], Faster R-CNN [15] and SPP [16]. The latter works in regression manner, whereby a single CNN is employed to directly predict the object category. Representative methods contain YOLOv2 [17], SSD [18], YOLOv3 [19], YOLOv4 [20] and YOLOv5.¹ Note that one-stage methods can obviously speed up detection, thereby facilitating real-time object monitoring and tracking.

Deep learning-based detectors trained in natural scenes would be prone to degrade in partial occlusion scenarios [21].

¹<https://github.com/ultralytics/yolov5>

Considering relationships between occluding and occluded objects, cases can be bisected into mutual occlusion between candidate objects and the one between candidate objects and others [22]. To be specific, as shown in Fig. 1, when the object is occluded by the same type, it is called intra-class occlusion, otherwise called inter-class occlusion [23]. Accordingly, main challenges are twofold: 1) Precise location of marine vessels is vitally deteriorated by occlusions. For instance, ships with similar characteristics occlude each other (see Fig. 1(a), (c) and (d)) are easily prone to be recognized as one target with a single prediction box; 2) Entire semantics of ships become rather weakly correlated to partial features. Specifically, there exist strong similarities between intra-class ships, such as the hulls of *Tanker* and *Containership*, and the masts of *Fishing* and *Sailingboat*.

To our best knowledge, there is limited research on occlusion detection of ships for the MASS. In [24], the mosaic data enhancement method has been deployed to augment training set of SeaShips [25] by using 4 images for cropping and stitching to form a new image, and has validated the ability to recognize overlapping ships. However, the synthetic data can not fully reflect actual occlusion scenarios. Notably, it is crucial to learn both local and global features under occluded situations. Attention mechanisms (AMs) have been widely applied to feature extraction since useful features can be enhanced while the rest can be suppressed. To be specific, the coordinate attention (CA) [26] has been extensively utilized in object detection in that position feature can be acquired to establish long-range feature dependencies. In [27], different-size filters have been used in front of corresponding CAs so as to capture multi-scale information, thereby obtaining the fusion of local and global features. It implies that the CA-based multi-receptive field position enhancement benefits the edge detection of changing targets. In [28], different-scale feature maps obtained from the SPP [16] and different-direction spatial feature maps obtained from the CA are flattened to enrich spatial features. To retain deep features of ships in IR images, the residual CA module (RCAM) has been proposed in [29] by using residual connection to combine a 2-D convolution layer with batch normalization, CA block, and rectified linear unit, thereby enhancing input feature maps and improving the representation of region of interest (RoI). Subsequently, in [30], the RCAM has been deployed to enhance spatial resolution of GF-1 WFV images, whereby deep high-frequency features can be extracted by deepening network layers using a residual-in-residual structure. Note that, as a plugin module, the CA can be conveniently embedded into deep learning models. For instance, inserting the CA into UNet [31] facilitates 2-D successive interpolation of the missing seismic data with a large gap. Incorporating the CA into CSPDarkNet53 [32] as a new backbone can promote recognition of surface defects of small printed circuit boards. Additionally, by inserting the CA into the last convolution layer of VGG16 [28], the classification of tuberculosis images can be empowered by capturing the cross-channel, direction and position sensing information.

Besides, the inference from partial visibility is essential for occlusion recognition, which attempts to establish semantics

correlation between local and global features. To this end, self-attention mechanism (SAM) [33] and image pyramids [34] have been deployed recently. To be specific, the self-attention transformer has been applied to capture target features appearing simultaneously on a feature map [12], thereby facilitating dense prediction task. Note that the transformer mechanism [35] can be integrated with feature pyramid transformer (FPT) [12] so as to capture cross-space and cross-scale feature interactions by transforming feature pyramid and performs better in selecting informative long-range interactions. However, expensive computations would be arisen in obtaining low-resolution feature maps and global self-attention. Accordingly, swin transformer (STR) with shifted windows [36] has been proposed by shrinking self-attention computation to non-overlapping local windows and allowing cross-window connections. Especially in dense detection tasks including classification and segmentation, the STR has achieved remarkable success [37].

Motivated by foregoing challenges and opportunities, in this paper, attention mechanisms within YOLO framework are innovated to accommodate accurate detection of occluded ships. Key contributions are summarized as follows:

- 1) A position enhancement module, termed RCAC3, is created by virtue of residual connections and the CAMEchanism extracting three-dimensional features of position, channel and space. In this context, the RCAC3 makes full use of shallow information and high-level semantics via residual connections, simultaneously, and establishes long-range feature dependencies, thereby facilitating accurate bounding boxes for occluded ships.
- 2) A multi-scale feature semantics correlation module, termed SP-STR, is devised by combining the SPP with the STR self-attention encoder block, so as to enhance the classification ability of occluded ships in both global and local sense.
- 3) The attention-aware occlusion detection scheme of maritime ships, termed AodeMar, is novelly developed by employing the RCAC3 and SP-STR modules to replace the convolution layers of backbone and neck networks, respectively. As a consequence, occlusion detection ability in accuracy and robustness can be sufficiently enhanced and can further be validated by establishing a real-world occlusion dataset.

The rest of this paper is organized as follows. Related works on occlusion detection are summarized in Section II. The proposed AodeMar framework is established in Section III. Experimental results and comparisons are presented in Section IV. Conclusions are drawn in Section V.

II. RELATED WORKS

A. Occlusion Detection Problem

Deep learning facilitates enormous developments of object detection, but is still limited in solutions to occlusion detection problem. For human beings, objects can be easily recognized even in the presence of significant occlusions, while deep learning-based detectors can hardly conduct foregoing task [21], [23]. According to proportion of

occlusion, i.e., 0%, 1%-35%, 35%-80% and 80%-100%, the cases can be classified as non-occlusion, general, severe and complete occlusions [22], which are in turn various occlusions are deployed to comprehensively evaluate detector performance [12], [25]. Actually, occlusion detection has been extensively studied in various tasks, e.g., pedestrian detection [38], [39], object tracking [40], face detection [41], vehicle detection [42], underwater image enhancement [43] and benthonic organism detection [44], *etc.*

In essence, deep learning-based detectors seriously rely on large-scale and informative datasets that can provide sufficient occlusion cases. It should be noted that general object-detection datasets, e.g., ImageNet [45], PASCAL VOC [46] and COCO [47], *etc.*, can not cover available occlusion images, making infeasible to train and/or evaluate occlusion detectors. Accordingly, specific occlusion datasets have been constructed, such as KITTI [48], Caltech Pedestrian [49], CityPersons [50], VehicleOcclusion [39] and CrowHuman [51], *etc.*

To our best knowledge, there is no publicly available marine vessel dataset exclusively for occlusion detection due to difficulties in obtaining samples within harsh marine environments. Alternatively, some existing marine vessel datasets have been deployed in occlusion detection problem. For instance, the SeaShips [25] has been employed to explore the impact of mosaic data enhancement strategy on overlapping occlusion of ships [24]. However, images are captured by onshore cameras and become unsuitable from the viewpoint of the MASS. In this paper, a customized dataset, named MVDD13 covering various occlusion scenarios, is deployed.

B. Occlusion Detection Method

Generally, marine vessel detection rarely takes into account the occlusion of ships which would be regarded as cases for robustness validation [12]. As mentioned in [25], it is difficult for detectors to distinguish 2 ships which are highly shaded from each other. Subsequently, in [24], mosaic data enhancement method has been deployed to augment training set by using 4 images for cropping and stitching to form a new image, which has been validated that the ability to recognize overlapping ships can be improved. However, the synthetic data can not fully reflect real-world occlusion scenarios. By extracting invariant features of ships, the co-occurrence histograms of oriented gradients [52] and constraint mutual subspace [53] have been utilized to tackle ship deformation due to partial loss. However, the resultant feature vectors become unsuitable for all ship types. To further solve occlusions problem during tracking process, blocking phenomenon judgment strategy and matching algorithm are usually combined to realize continuous tracking after blocking [53], [54]. It should be noted that foregoing methods become unavailable for detecting occluded objects in static images.

Observed from pedestrian, crowd and vehicle detection, and pavement damage detection tasks, *etc.*, occlusion detection methods can be typically devised by data enhancement,

semantic segmentation and network structure design, respectively. From the respect of data enhancement, noises are added into raw images, thereby simulating occlusions. Usually, occlusion operations including Cutout [55], CutMix [56] and Mosaic [24], *etc.*, can alleviate the occlusion to some extent, but can not capture occlusion features. Alternatively, semantic segmentation can achieve target classification and detection through dense inference at pixel level. To this end, detection-based video object segmentation [57], [58], [59] was achieved by matching the target appearance given by the first frame to subsequent frames, thereby robust to occlusion and drift, although the algorithms essentially require object continuity and mask labels. From the viewpoint of network structure optimization, detection accuracy of occluded objects can be enhanced by strengthening feature expression ability (i.e., RoI pooling [38], [60]) and improving regression loss of prediction boxes (i.e., repulsion [61] and aggregation [38] losses, IoU loss [62], [63] and non-maximum suppression (NMS) [64], [65]). Although those methods can alleviate the overlapping of boundary boxes, the essence is that all highly overlapping prediction boxes are taken as false positives.

Subsequently, the spatial and channel direction information has been explored to enhance feature representation. For instance, the squeeze-and-excitation network (SE) [66], convolutional block attention module (CBAM) [67] and CA [26], *etc.*, aim to extract key information from a vast amount of data and allocate more attention to useful resources. Recently, plug-and-play attention modules have been incorporated into various object detection models [27], [28], [29], [30]. Different from AMs, the core of SAMs is to capture the correlations between features. Inspired by successes in NLP, multiple SAMs including vision transformer [35], [68] and STR [36] have been extensively employed to capture features that appear simultaneously in the same scene.

In summary, occlusion detection of ships is still open for active exploration and development. It is highly expected to borrow powerful paradigms from inland scenarios, so as to pave an efficient path to occlusion detection for marine ships.

III. AOEDMAR FRAMEWORK

In this section, the proposed AodeMar framework shown in Fig. 2 is established by devising two modules, i.e., the position enhancement module and the multi-scale feature semantics correlation module.

A. Position Enhancement Module

In the scenario of densely distributed multi-targets, marine vessels tend to be mutually occluded by each other. Even worse, high similarities between marine vessels make it difficult to distinguish target edges in the occluded area, thereby leading to imprecise or unable detection. In this context, key information with discriminatory attributes are required to be enhanced, while redundant information has to be weakened. Moreover, position information of target edges occluded is expected to be paid additional attention.

Inspired by [26], [29], and [30], a position enhancement module is ingeniously designed. As shown in Fig. 3,

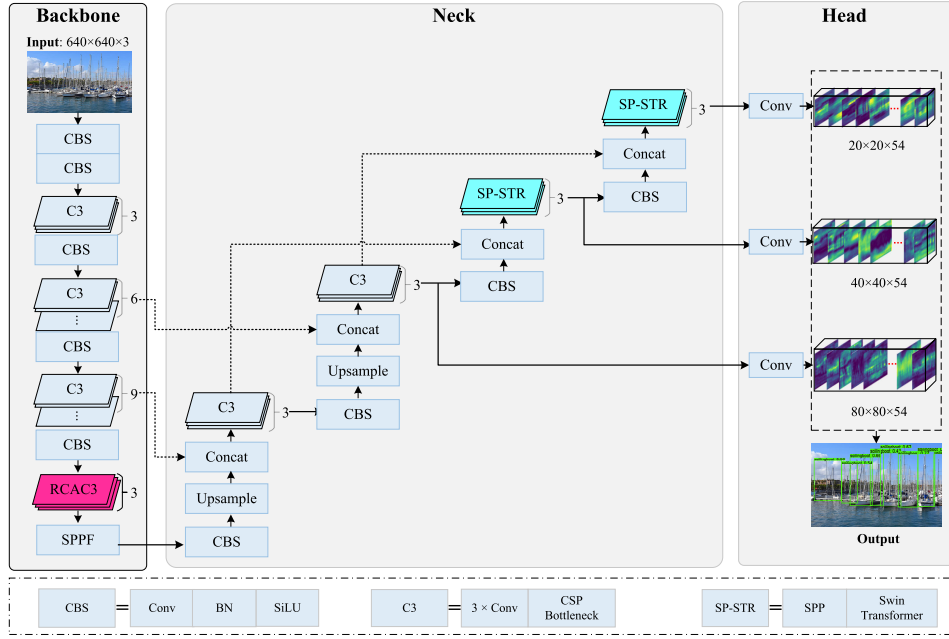


Fig. 2. The overall architecture of AodeMar framework.

a residual coordinate attention block (RCAB) is devised to fuse shallow- and high-level semantics together with spatial feature dependencies. Furthermore, to further enhance feature extraction ability of C3 module, the RCAB is incorporated into it to form RCAC3 module.

The input of the CA is $\mathcal{X}_C = [X_1, X_2, \dots, X_c] \in \mathbb{R}^{c \times m \times n}$, where c , m and n denote the number of channels and dimensions in height and width directions, respectively, $X_k \in \mathbb{R}^{m \times n}$ denotes the k -th channel feature map. Firstly, to embed the coordinate information, 2 average pooling kernels of size $m \times 1$ and $1 \times n$ are employed to encode each channel along the horizontal and vertical coordinate directions, respectively. Subsequently, the output of the k -th channel at height h and width w can be expressed by

$$Z_k^h = \frac{1}{n} \sum_{i=0}^{n-1} X_k(m, i) \quad (1)$$

$$Z_k^w = \frac{1}{m} \sum_{j=0}^{m-1} X_k(j, n) \quad (2)$$

where $Z_k^h \in \mathbb{R}^{m \times 1}$ and $Z_k^w \in \mathbb{R}^{1 \times n}$, $k = 1, 2, \dots, c$ and $(\cdot, \cdot)$ denotes the position.

It is followed by coordinate attention generation process, whereby feature maps Z_k^h and Z_k^w are concatenated and are sent to a shared 1×1 convolution transformation function F . Using a nonlinear activation function δ , the spatial information can be encoded in feature maps given by

$$G_l = \delta(F([Z_k^h]^T, Z_k^w)) \quad (3)$$

where $G_l \in \mathbb{R}^{1 \times (m+n)}$, $l = 1, 2, \dots, c/r$ and r is a reduction factor, $(\cdot)^T$ and $[\cdot, \cdot]$ denote transposition and concatenation operations along spatial dimension, respectively.

Afterwards, the corresponding feature maps are generated by splitting G_l into G_l^h and G_l^w along position directions.

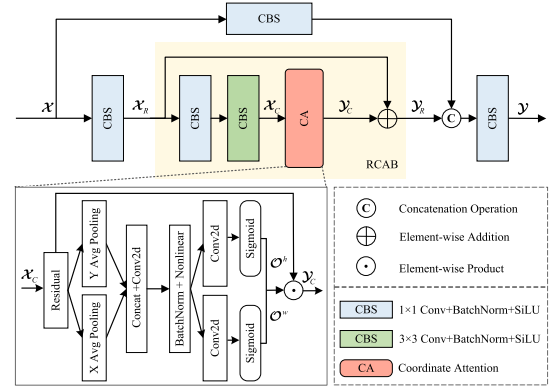


Fig. 3. The architecture of position enhancement module.

Subsequently, 1×1 convolution functions L_h and L_w are employed to separately transform $\mathcal{G}^h = [G_1^h, G_2^h, \dots, G_{c/r}^h] \in \mathbb{R}^{c/r \times m \times 1}$ and $\mathcal{G}^w = [G_1^w, G_2^w, \dots, G_{c/r}^w] \in \mathbb{R}^{c/r \times 1 \times n}$ to tensors with the same channels to the input $\mathcal{X}_C$, i.e., Eqn. 8. The outputs with precise coordinate information can be computed by

$$O_k^h = \gamma(L_h(G_l^h)) \quad (4)$$

$$O_k^w = \gamma(L_w(G_l^w)) \quad (5)$$

where γ is Sigmoid function, $O_k^h \in \mathbb{R}^{m \times 1}$ and $O_k^w \in \mathbb{R}^{1 \times n}$.

Eventually, O_k^h and O_k^w are applied as position attention weights to obtain the output of CA as follows:

$$\mathcal{Y}_C = \mathcal{O}^h \otimes \mathcal{O}^w \odot \mathcal{X}_C \quad (6)$$

where “ $\otimes$ ” and “ $\odot$ ” denote tensor product and element-wise product operations, respectively, $\mathcal{Y}_C = [Y_1, Y_2, \dots, Y_c] \in \mathbb{R}^{c \times m \times n}$, $\mathcal{O}^h = [O_1^h, O_2^h, \dots, O_{c/r}^h] \in \mathbb{R}^{c \times m \times 1}$ and $\mathcal{O}^w = [O_1^w, O_2^w, \dots, O_{c/r}^w] \in \mathbb{R}^{c \times 1 \times n}$.

As shown in Fig. 3, $\mathcal{X} \in \mathbb{R}^{c \times m \times n}$ and $\mathcal{Y} \in \mathbb{R}^{c \times m \times n}$ denote input and output feature maps of the RCAC3, respectively.

Note that the RCAC3 is formed by inserting the RCAB block into C3 module (i.e., 3 CBS in addition to RCAB in Fig. 3), so as to further enhance the feature extraction ability of backbone. In this context, the input of RCAB $\mathcal{X}_R \in \mathbb{R}^{c \times m \times n}$ can be calculated by

$$\mathcal{X}_R = \mathcal{F}_1(\mathcal{X}) \quad (7)$$

where $\mathcal{F}_1$ is CBS operation with 1×1 convolution kernel.

Subsequently, in RCAB block, 2 CBS modules including 1×1 and 3×3 convolution operations, batch normalization (BatchNorm) and sigmoid linear unit (SiLU) activation function are employed before the CA in sequence, and a residual connection is added to fuse the low- and high-level feature information. In this context, the output of the RCAB $\mathcal{Y}_R \in \mathbb{R}^{c \times m \times n}$ can be calculated by

$$\mathcal{X}_C = \mathcal{F}_3(\mathcal{F}_1(\mathcal{X}_R)) \quad (8)$$

$$\mathcal{Y}_R = \mathcal{Y}_C \oplus (\mathcal{K} \odot \mathcal{X}_R) \quad (9)$$

where $\mathcal{F}_3$ denotes CBS operation with 3×3 convolution kernel, $\mathcal{K}$ is a coefficient matrix, whereby each element belongs to $[0, 1]$, “ $\oplus$ ” denotes element-wise addition operation.

Finally, the output of the position enhancement module RCAC3 can be expressed by

$$\mathcal{Y} = \mathcal{F}_1(\mathcal{Y}_R \odot \mathcal{F}_1(\mathcal{X})) \quad (10)$$

where “ $\odot$ ” is concatenation operation in channel dimension.

Remark 1: From (1) and (2), we can observe that by decomposing global pooling into 2 separate average pooling operations along height and width directions, the precise position information can be preserved and is utilized to capture long-range dependencies along spatial direction, so as to accurately locate the objects of interest.

Remark 2: The concatenation in spatial dimension in (3) can effectively capture inter-channel relationships [66], [67], while that in channel dimension in (10) can capture inter-spatial relationships. Together with channel, spatial and position information, the RCAC3 module can be beneficial to feature representation.

B. Multi-Scale Feature Semantics Correlation Module

Note that the entirety semantics of a ship are weakly correlated with partial features. It implies that invisible occlusion can hardly be inferred from unoccluded features. Nevertheless, the intrinsic locality of CNN-based approaches impedes learning the interactions between global and local semantics [69].

Inspired by [12], [36], and [37], a multi-scale feature semantics correlation module, termed SP-STR, is devised by combining the SPP with STR, so as to ensure the classification ability in entirety and parts.

As shown in Fig. 4, the input feature map of the SP-STR is $\mathcal{X}_S \in \mathbb{R}^{c_s \times m_s \times n_s}$, where c_s , m_s and n_s denote the number of channels and dimensions at height and width directions, respectively. Firstly, CBS block with 1×1 convolution operation is employed to encode each channel of input feature map and is given by

$$\mathcal{X}_{S'} = \mathcal{F}_1(\mathcal{X}_S) \quad (11)$$

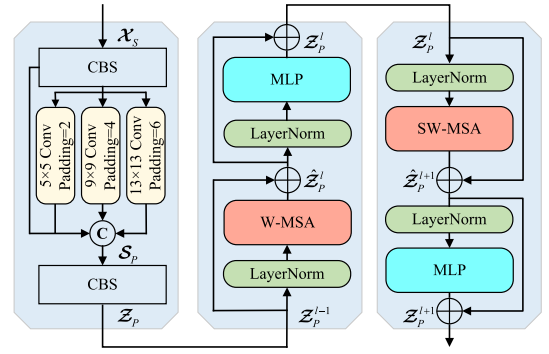


Fig. 4. The architecture of multi-scale feature semantics correlation module.

where $\mathcal{X}_{S'} \in \mathbb{R}^{(c_s/2) \times m_s \times n_s}$, $\mathcal{F}_1$ denotes CBS operation with 1×1 convolution kernel.

Followed by max-pooling layers with kernel sizes of 5×5 , 9×9 and 13×13 , respectively, the concatenated operation is implemented for fusing the global and local feature information, and can be defined by

$$\mathcal{S}_{k_s} = \mathcal{P}_{k_s}(\mathcal{X}_{S'}), \quad k_s = 5, 9, 13 \quad (12)$$

$$\mathcal{S}_P = \mathcal{S}_1 \odot \mathcal{S}_5 \odot \mathcal{S}_9 \odot \mathcal{S}_{13} \quad (13)$$

where $\mathcal{P}_{k_s}$ denotes max-pooling operations with different pooling kernel sizes (i.e., $k_s = 5, 9, 13$) and the corresponding feature map $\mathcal{S}_{k_s} \in \mathbb{R}^{(c_s/2) \times m_s \times n_s}$ is extracted. Note that $\mathcal{S}_1 = \mathcal{X}_{S'}$, $\mathcal{S}_P \in \mathbb{R}^{(2c_s) \times m_s \times n_s}$ denotes the concatenated feature map.

The output channel of the SPP is restored by the CBS block with 1×1 convolution operation, which is expressed by

$$\mathcal{Z}_P = \mathcal{F}_1(\mathcal{S}_P) \quad (14)$$

where $\mathcal{Z}_P \in \mathbb{R}^{c_s \times m_p \times n_p}$.

Subsequently, $\mathcal{Z}_P$ containing multi-scale feature information is input of 2 consecutive STR self-attention [36] blocks, whereby each contains 2 sub-layers, i.e., windows-based multi-head self attention (W-MSA) and multi-layer perceptron (MLP) layers. Specifically, layer normalization (LayerNorm) is employed to normalize different features of a single sample in channel dimensions. Accordingly, W-MSA is utilized to construct information correlations in the window. To mitigate vanishing gradient, a residual connection combines the forward entire information to the layer for local-global semantic feature learning. The output of the sub-layer for l -th block can be expressed by

$$\hat{\mathcal{Z}}_P^l = \mathcal{F}_W(\mathcal{L}_N(\mathcal{Z}_P^{l-1})) \oplus \mathcal{Z}_P^{l-1} \quad (15)$$

where $\hat{\mathcal{Z}}_P^l \in \mathbb{R}^{c_s \times m_p \times n_p}$, $\mathcal{F}_W$ and $\mathcal{L}_N$ denote W-MSA and LayerNorm operations, respectively. Note that $\mathcal{Z}_P^{l-1} = \mathcal{Z}_P$.

Afterwards, MLP layer is deployed to learn feature weights for global perception, and can be given by

$$\mathcal{Z}_P^l = \mathcal{F}_M(\mathcal{L}_N(\hat{\mathcal{Z}}_P^l)) \oplus \hat{\mathcal{Z}}_P^l \quad (16)$$

where $\mathcal{Z}_P^l \in \mathbb{R}^{c_s \times m_p \times n_p}$, $\mathcal{F}_M$ denotes MLP operation.

To further reduce computations of MSA and establish the information connections across windows. Similarly, the shifted

Algorithm 1 Entire AodeMar Algorithm

```

1 /* Set the training stage */
   Input: Training images with annotations;
   Output: Optimized AodeMar model;
2 number of epochs  $n^e \leq 300$ ;
3 number of batch size  $n^b = 32$ ;
4 for  $i \leftarrow 1$  to  $n^e$  do
5   for  $j \leftarrow 1$  to  $n^b$  do
6     /* Position enhancement module */
7     /* Calculate output of RCAB:  $\mathcal{Y}_R$  */
8      $\mathcal{X}_R = \mathcal{F}_1(\mathcal{X}) \leftarrow \text{eqn. (7)}$ 
9      $\mathcal{X}_C = \mathcal{F}_3(\mathcal{F}_1(\mathcal{X}_R)) \leftarrow \text{eqn. (8)}$ 
10     $\mathcal{Y}_C = \mathcal{O}^h \otimes \mathcal{O}^w \odot \mathcal{X}_C \leftarrow \text{eqn. (6)}$ 
11     $\mathcal{Y}_R = \mathcal{Y}_C \oplus (\mathcal{K} \odot \mathcal{X}_R) \leftarrow \text{eqn. (9)}$ 
12    /* Calculate output of module:  $\mathcal{Y}$  */
13     $\mathcal{Y} = \mathcal{F}_1(\mathcal{Y}_R \odot \mathcal{F}_1(\mathcal{X})) \leftarrow \text{eqn. (10)}$ 
14    /* Multi-scale feature semantics correlation
       module */
15    /* Calculate  $\mathcal{S}_P$  */
16     $\mathcal{X}_{S'} = \mathcal{F}_1(\mathcal{X}_S) \leftarrow \text{eqn. (11)}$ 
17     $\mathcal{S}_{k_s} = \mathcal{P}_{k_s}(\mathcal{X}_{S'})$ ,  $k_s = 5, 9, 13 \leftarrow \text{eqn. (12)}$ 
18     $\mathcal{S}_P = \mathcal{X}_{S'} \odot \mathcal{S}_5 \odot \mathcal{S}_9 \odot \mathcal{S}_{13} \leftarrow \text{eqn. (13)}$ 
19    /* Calculate  $\mathcal{Z}_P^{l-1}$  */
20     $\mathcal{Z}_P = \mathcal{F}_1(\mathcal{S}_P) \leftarrow \text{eqn. (14)}$ 
21     $\mathcal{Z}_P^{l-1} = \mathcal{Z}_P$ ;
22    /* Calculate output of module:  $\mathcal{Z}_P^{l+1}$  */
23     $\hat{\mathcal{Z}}_P^l = \mathcal{F}_W(\mathcal{L}_N(\mathcal{Z}_P^{l-1})) + \mathcal{Z}_P^{l-1} \leftarrow \text{eqn. (15)}$ 
24     $\mathcal{Z}_P^l = \mathcal{F}_M(\mathcal{L}_N(\hat{\mathcal{Z}}_P^l)) + \hat{\mathcal{Z}}_P^l \leftarrow \text{eqn. (16)}$ 
25     $\hat{\mathcal{Z}}_P^{l+1} = \mathcal{F}_S(\mathcal{L}_N(\mathcal{Z}_P^l)) + \mathcal{Z}_P^l \leftarrow \text{eqn. (17)}$ 
26     $\mathcal{Z}_P^{l+1} = \mathcal{F}_M(\mathcal{L}_N(\hat{\mathcal{Z}}_P^{l+1})) + \hat{\mathcal{Z}}_P^{l+1} \leftarrow \text{eqn. (18)}$ 
27   end
28 end

```

W-MSA (SW-MSA) and responding MLP are computed by

$$\hat{\mathcal{Z}}_P^{l+1} = \mathcal{F}_S(\mathcal{L}_N(\mathcal{Z}_P^l)) \oplus \mathcal{Z}_P^l \quad (17)$$

$$\mathcal{Z}_P^{l+1} = \mathcal{F}_M(\mathcal{L}_N(\hat{\mathcal{Z}}_P^{l+1})) \oplus \hat{\mathcal{Z}}_P^{l+1} \quad (18)$$

where $\hat{\mathcal{Z}}_P^{l+1} \in \mathbb{R}^{c_s \times m_p \times n_p}$ and $\mathcal{Z}_P^{l+1} \in \mathbb{R}^{c_s \times m_p \times n_p}$, $\mathcal{F}_S$ denotes SW-MSA operation.

Remark 3: The semantics correlation information is developed by the shifted windows based self-attention operation in a connections-across-windows manner. Specifically, correlations between neighboring non-overlapping windows are created such that semantics between local and global features can be comprehensively obtained.

Remark 4: The hierarchical architecture of SP-STR provides flexibility for modeling at various scales and has linear computational complexity with respect to image sizes, such that the proposed SP-STR is able to replace convolution operations and can learn semantics correlation between feature blocks with different attentions.

TABLE I
NUMBER OF OBJECTS IN EACH CATEGORY

Category	Objects	Percentage
Cargo	7,640	0.1871
Passenger	5,685	0.1392
Cruise	5,307	0.1299
Bulker	5,270	0.1290
Tanker	4,589	0.1124
Sailingboat	3,830	0.0938
Tug	2,733	0.0669
Fishing	1,427	0.0349
Drill	1,336	0.0327
Firefighting	1,278	0.0313
Containership	1,074	0.0263
Warship	420	0.0103
Submarine	250	0.0061

TABLE II
AODEMAR HYPERPARAMETERS

Learning rate l_r	Momentum m	Weight decay w_d	mAP@.5 (%)
0.01	0.90	0.0001	94.65
0.002	0.90	0.0001	94.86
0.0001	0.90	0.0001	94.23
0.002	0.90	0.0009	95.02
0.002	0.90	0.0005	95.43
0.002	0.95	0.0005	95.35
0.002	0.85	0.0005	95.23

Remark 5: As demonstrated in [70], feature maps at the end of network are low resolution, thereby reducing computation and memory cost. In addition, experimental results show that replacing the last 2 convolution operations in the neck network with the SP-STR modules can benefit detection accuracy.

As summarized in **Algorithm 1**, the entire AodeMar scheme is established. Specifically, the RCAC3 is designed to make full use of shallow information and high-level semantics via residual connections, and focuses on the position information of objects and establishes long-distance feature dependencies with the aid of CA, so as to improve the localization ability of model. Consequently, the SP-STR is devised by combining the SPP with STR self-attention block to capture multi-scale semantics correlation in both global and local sense, thereby enhancing the classification ability of occluded ships. Moreover, the RCAC3 is employed in the last convolution layer of backbone, while 2 SP-STR modules are applied to respectively replace the last 2 convolution layers of neck network.

IV. EXPERIMENTAL STUDIES AND COMPARISONS

Setups including dataset, evaluation metrics and experimental configuration are firstly introduced. In order to demonstrate the effectiveness and superiority of the AodeMar, ablation studies, quantitative comparisons, real-world and robustness performance validation experiments are presented in details.

A. Experimental Settings

1) *Dataset:* There are few publicly available datasets of marine vessels detection in overlapping occlusion for the

TABLE III
ABLATION STUDIES IN DIFFERENT MODELS

		Frameworks									AodeMar (Our)
Modules	Baseline	✓									
	RSEC3		✓						✓		
	RCBAMC3			✓						✓	
	RCAC3				✓						✓
	STR					✓		✓			
	SP-STR						✓		✓		
mAP@.5	–	94.08	94.97	95.02	<u>95.23</u>	94.84	94.87	94.96	94.98	94.71	95.43
mAP@[.5:.95]	–	79.25	82.28	81.77	<u>82.43</u>	81.04	81.00	<u>82.52</u>	81.61	81.05	82.57
AP (%)	Cargo	97.12	97.13	96.98	96.96	97.21	97.06	97.16	97.19	<u>97.20</u>	97.25
	Passenger	94.49	94.27	94.14	93.80	94.32	94.22	94.36	<u>94.42</u>	94.17	94.05
	Cruise	<u>98.63</u>	97.73	98.55	98.66	98.31	98.11	98.48	97.69	98.77	98.11
	Bulker	98.58	98.77	98.81	98.66	98.62	98.89	98.67	98.58	<u>98.87</u>	98.78
	Tanker	97.87	98.20	98.53	98.53	98.49	98.61	98.67	98.68	98.85	<u>98.77</u>
	Sailingboat	95.62	93.89	94.14	<u>95.12</u>	94.02	94.00	93.72	93.57	92.65	94.33
	Tug	88.36	<u>89.52</u>	89.55	88.93	89.44	88.25	88.10	88.20	87.42	88.23
	Fishing	79.64	85.12	83.66	85.52	83.33	<u>86.45</u>	84.61	86.17	85.26	88.61
	Drill	98.53	98.29	98.08	98.29	97.48	98.21	99.20	97.55	98.89	<u>98.95</u>
	Firefighting	89.77	93.55	94.82	93.92	93.54	92.99	94.46	94.01	93.39	93.74
	Containership	91.13	93.32	93.61	<u>94.24</u>	94.39	92.99	91.44	93.14	91.15	93.44
	Warship	96.68	96.38	95.58	96.56	94.50	96.15	96.10	97.11	95.92	96.78
	Submarine	96.57	98.40	97.83	98.73	99.47	97.35	99.50	98.40	98.64	99.50

Note: The values in bold and underline in each row denote the best and sub-optimal AP, mAP@.5 and mAP@[.5:.95], respectively.

MASS. In our previous work, a marine vessel detection dataset, named MVDD13² [71], has been established and publicly available, which contains 13 categories with precise annotations, i.e., *Cargo*, *Passenger*, *Cruise*, *Bulker*, *Tanker*, *Sailingboat*, *Tug*, *Fishing*, *Drill*, *Firefighting*, *Containership*, *Warship* and *Submarine*. Note that various degrees of occlusion conditions are included in the MVDD13, thereby making suitable for occlusion detection evaluation in this paper. The sample distribution of MVDD13 is shown in Table I. In all experiments, 25,541, 2,838 and 7,095 images are used for training, validation and testing, respectively.

2) *Evaluation Metrics*: The indices including precision (Pr), recall (Re), F_1 , average precision (AP) for single category, mean AP (mAP) and frames per second (FPS) are deployed as evaluation metrics. Given an IoU threshold, the Pr and Re can be determined accordingly. To this end, the AP, i.e., the area of precision-recall (P-R) curve, can be calculated by

$$AP = \int_0^1 Pr(Re) dRe \quad (19)$$

Accordingly, the average detection accuracy of 13 categories in various IoU thresholds, i.e., mAP@.5 and mAP@[.5:.95], can be obtained. In addition, the FPS³ can be calculated directly by the number of frames divided by time cost.

3) *Configuration Details*: All experiments are conducted on a Linux PC platform with an Intel Core i7-8700K CPU and an NVIDIA TITAN Xp (12GB RAM). In addition, both training and testing environments are built on Ubuntu 18.04 operating system running CUDA 10.2 and cuDNN v8.3.2.

²<https://github.com/yuanwang1010/MVDD13>

³<https://mmdetection.readthedocs.io/zhCN/latest/userguides/usefultools.html?highlight=fps>

Note that the learning rate l_r affects convergence and weight decay w_d is used to prevent overfitting, while the momentum m is tuned to speed up the stochastic gradient descent (SGD) algorithm. Accordingly, to properly determine hyperparameters, representative selections are summarized in Table II, and are evaluated in terms of mAP@.5. It shows that the suite of combination, i.e., $l_r = 0.002$, $m = 0.9$ and $w_d = 0.0005$ preserves the best performance. In summary, a moderate choice of learning rate renders smooth convergence without trapping in local minima. Using a slightly large momentum, faster speed towards the global minimum can be obtained. A moderately small weight decay is apt to prevent overfitting, thereby enhancing generalization.

B. Ablation Studies

In order to demonstrate the superiority of the AodeMar scheme, the AP, mAP@.5 and mAP@[.5:.95] are used as quantitative measures. Accordingly, the impact of each component is listed in Table III.

1) *Effect of the RCAC3 Module*: Compared to the baseline, the use of RCAC3 module does not increase the total number of layers in the model, and further results in improvements of 1.15 % and 3.18% in mAP@.5 and mAP@[.5:.95], respectively. Additionally, compared with residual connections-based SE and CBAM AMs (referred to RSEC3 and RCBAMC3), the RCAC3 exhibits significantly superiority in both mAP@.5 and mAP@[.5:.95] since the location information is embedded into channel attention. It is illustrated that the CA outperforms in enhancing the feature extraction ability of the backbone network compared to the

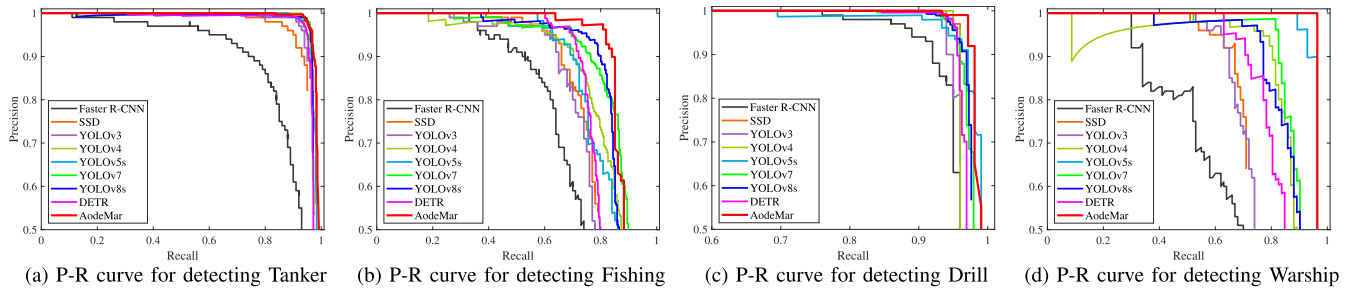


Fig. 5. Detection performance comparisons on Tanker, Fishing, Drill and Warship types.

TABLE IV
COMPARISONS WITH TYPICAL DETECTORS

Metrics		Methods								
		Faster R-CNN	SSD	YOLOv3	YOLOv4	YOLOv5s	YOLOv7	YOLOv8s	DETR	AodeMar
mAP@.5		80.36	88.47	89.20	92.68	94.08	94.05	93.54	90.21	95.43
mAP@[.5:.95]		51.60	67.57	66.37	66.93	79.25	81.50	81.52	77.92	82.57
Pr		85.51	93.15	93.50	94.86	94.61	92.52	93.35	81.38	95.02
Re		81.21	81.28	84.63	83.37	88.89	90.70	89.97	89.87	91.02
F ₁		83.30	86.81	88.84	88.74	91.66	91.60	91.63	85.41	92.98
Parameter (M)		33.04	26.29	61.95	64.36	7.06	37.62	11.17	36.76	8.28
FLOPs (G)		134.38	62.75	66.17	60.53	16.71	106.47	28.82	114.25	67.40
FPS		1.21	38.76	25.49	21.12	127.88	52.72	116.42	33.82	99.01
AP (%)	Cargo	92.72	96.22	97.60	98.07	97.12	96.22	96.05	95.42	97.25
	Passenger	87.86	90.18	92.81	94.73	94.49	94.65	94.11	91.65	94.05
	Cruise	95.97	97.09	98.15	98.61	98.63	98.06	98.37	97.01	98.11
	Bulker	90.86	96.56	97.16	98.51	98.58	95.73	95.54	94.23	98.78
	Tanker	87.61	94.54	96.68	98.03	97.87	96.89	96.64	95.72	98.77
	Sailingboat	85.08	87.85	91.19	93.20	95.62	95.30	94.75	90.67	94.33
	Tug	68.62	83.71	85.63	90.61	88.36	90.22	89.73	83.89	88.23
	Fishing	70.07	75.55	76.22	83.25	79.64	86.36	84.94	82.35	88.61
	Drill	94.65	96.39	95.41	96.22	98.53	97.52	97.28	95.74	98.95
	Firefighting	69.80	89.35	92.11	96.43	89.77	93.72	94.24	87.77	93.74
	Containership	72.26	83.46	85.77	89.55	91.13	91.03	94.05	84.39	93.44
	Warship	62.31	69.45	71.74	76.39	96.68	89.70	86.76	83.15	96.78
	Submarine	66.84	89.82	79.13	91.19	96.57	97.24	96.70	91.48	99.50

Note: The values in bold in each row denote the best mAPs, Pr, Re, F₁, parameters, FLOPs, FPS and AP, respectively.

TABLE V
INTENSIVE COMPARISONS IN OCCLUDED AND UNOCCLUDED CASES

Test data	Metrics	Methods					
		YOLOv4	YOLOv5s	YOLOv7	YOLOv8s	DETR	AodeMar
UI	mAP@.5	93.11	93.34	94.49	93.75	92.78	94.01
	mAP@[.5:.95]	72.91	78.77	69.44	78.55	71.06	78.97
	US (6996)	5655	5949	5876	5988	5900	5993
OI	mAP@.5	68.09	69.94	69.01	70.14	66.58	71.13
	mAP@[.5:.95]	44.31	49.62	52.50	53.24	50.24	52.56
	OS (1173)	662	693	697	700	679	738

typical SE and CBAM AMs, since the CA can capture three-dimensions information including channel, spatial and position. Especially, obtained position attention information helps to establish long-distance feature dependencies, thereby facilitating accurate bounding boxes (BBboxes) for occluded ships.

2) *Effect of the SP-STR Module*: Compared to the baseline network, the use of STR blocks results in improvements of 0.76% and 1.79% in mAP@.5 and mAP@[.5:.95], respectively, since its hierarchical structure with 2 successive

sub-layers and cross-windows connections can enhance the feature semantics correlation of different-layer local-global information. Furthermore, by integrating the SPP into the STR self-attention block, the SP-STR achieves a higher mAP@.5, indicating that the SPP is beneficial to detect occluded ships by providing multi-scale feature semantics. Note that combined with RCAC3 module, the SP-STR is remarkably superior to STR not only in mAP@.5 and mAP@[.5:.95] but also in most of APs, especially for densely appearing ship types including *Fishing*, *Warship* and *Sailingboat*. It eventually implies that the SP-STR significantly enhances the detection performance of occluded ships.

3) *Effect of the AodeMar Framework*: The AodeMar framework, which combines the RCAC3 and SP-STR modules, exhibits significant superiority in terms of mAP@.5 and mAP@[.5:.95]. Although not all categories show the best APs, optimal and sub-optimal APs are obtained in most categories. In particular, the AP performance of categories including *Fishing*, *Firefighting* and *Containership* which are prone to occlusion is significantly improved by 8.97%, 3.97%

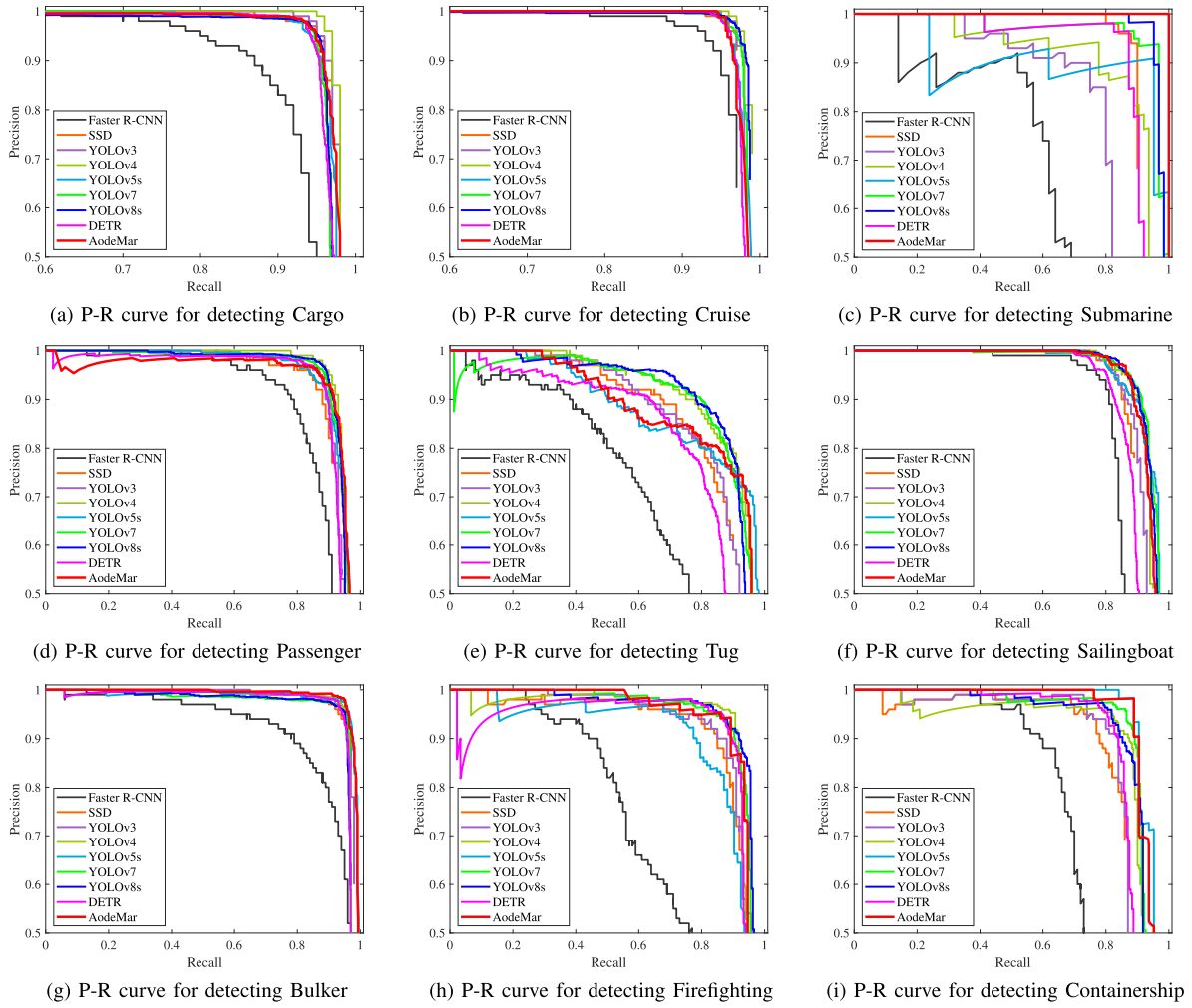


Fig. 6. Detection performance comparisons on Cargo, Cruise, Submarine, Passenger, Tug, Sailingboat, Bulker, Firefighting and Containership types.

and 2.93%, respectively. As a consequence, the AodeMar framework own superior detection performance.

C. Quantitative Comparisons

To sufficiently demonstrate the superiority of the AodeMar, the typical two-stage Faster R-CNN [15], one-stage approaches, e.g., SSD [18], YOLOv3 [19], YOLOv4 [20], YOLOv5s, YOLOv7 [72] and YOLOv8s, and attention-based DETR [73] are considered for comparisons in detection accuracy and speed. It should be noted that, for fair comparisons, aforementioned schemes are thoroughly pre-trained on COCO dataset and fine-tuned on MVDD13 dataset.

Accordingly, comparison results are listed in Table IV, from which we can see that the AodeMar achieves much higher mAP@.5, mAP@[.5:.95], Pr, Re and F₁, since the location and recognition ability can be promoted by virtue of position enhancement and semantics correlation modules. Note that, besides Pr and Re, comprehensive metrics mAPs and F₁ can evaluate the overall performance. Specifically, the AodeMar performs significant improvements with at least 1.43% mAP@.5 (i.e., from 94.08% to 95.43%), 1.29% mAP@[.5:.95] (i.e., from 81.52% to 82.57%), and 1.37% F₁ (i.e., from 91.66% to 92.98%). With regard to detection speed, the AodeMar performs 99.01FPS, using 8.28M parameters and

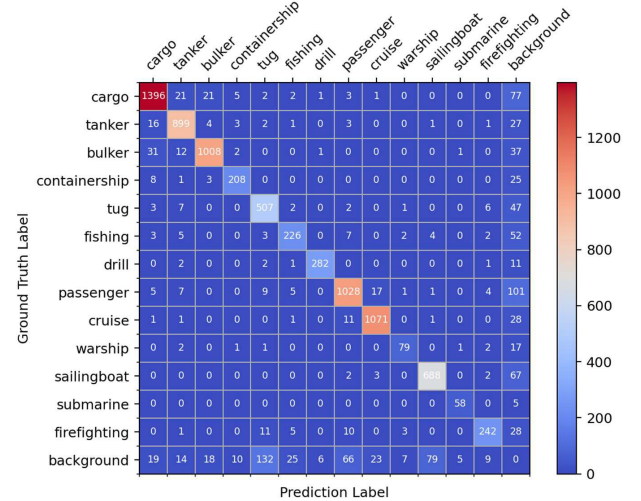


Fig. 7. Confusion matrix from AodeMar on MVDD13 test set, with IoU and confidence thresholds of 0.5 and 0.25, separately.

67.4G FLOPs and is clearly superior to counterparts except for the YOLOv5s and YOLOv8s. It should be noted that the two-stage method, i.e., Faster R-CNN, becomes inferior in both accuracy and speed due to the lack of semantics for small-scale occluded ships and time-consuming selective search method.

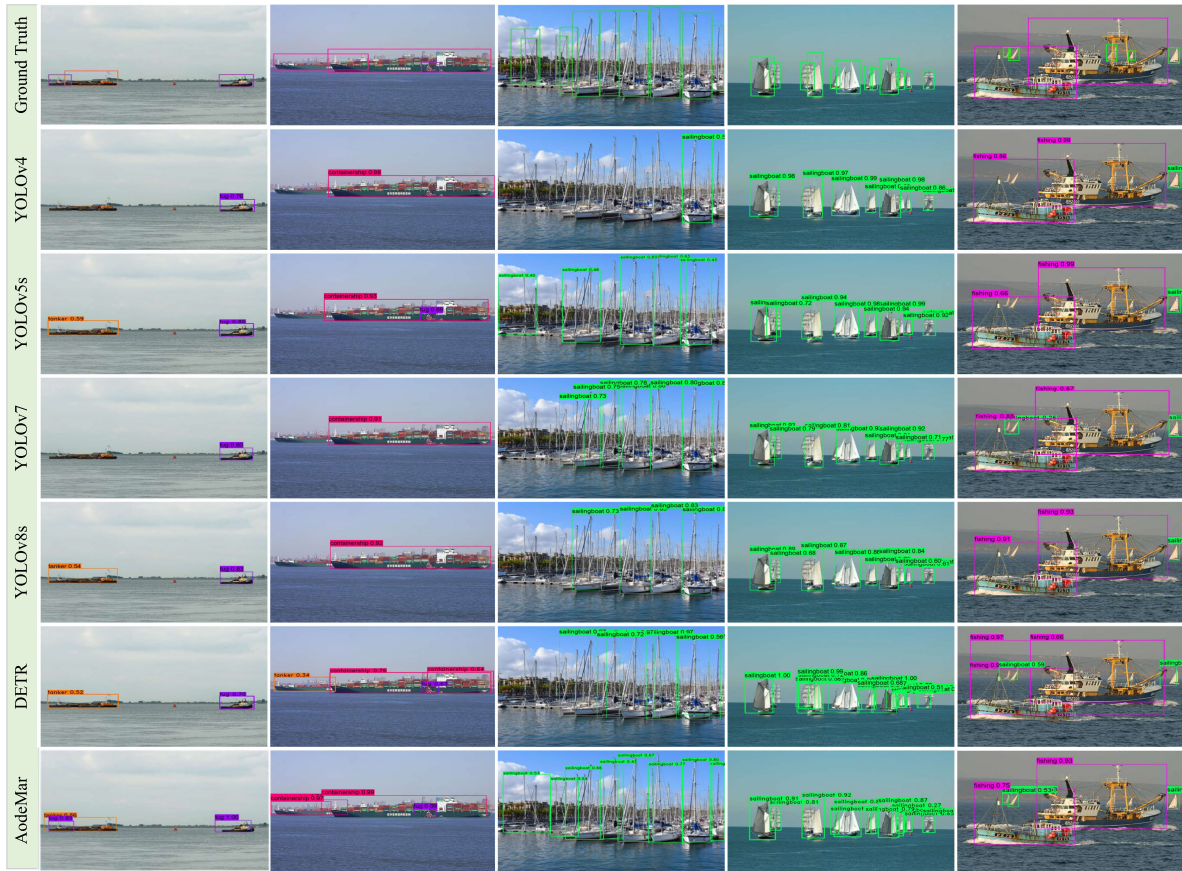


Fig. 8. The detection results in various occlusion scenarios.

By comparisons, the YOLOv4 and YOLOv5s perform partial superiority in some ship types since Mosaic and SPP modules enable detection enhancement. Comparing to the YOLOv5s with the highest FPS, the AodeMar using RCAC3 and SP-STR modules showcases remarkable increases in both $mAP@.5$ and $mAP@[.5:.95]$. Especially comparing to the YOLOv8s with anchor-free lightweight, the AodeMar uniformly provides higher accuracy while the detection speed would be slightly sacrificed. Fortunately, the FPS executed by the AodeMar is apparently competent for a real-time detection task.

In terms of precision and recall balance, P-R curves with 0.5 IoU threshold for each category are provided in Figs. 5 and 6. Obviously, the AodeMar outperforms in most of ship types including *Tanker*, *Fishing*, *Drill*, *Warship*, *Submarine* and *Bulker*. For *Cargo*, *Passenger* and *Containership*, the AodeMar exhibits much higher recall rate whenever the precision tends to increase. It implies that target ships can be sorted out to the utmost extent provided that those objects can be detected, thereby beneficially supporting MASS navigation safety. In terms of *Sailingboat*, *Cruise* and *Firefighting* categories, most of approaches are able to make satisfactory detection. As for the *Tug*, the AodeMar is good at recognizing false negative instances, but suffers from strong interference between precision and recall especially around the break even point (BEP). Moreover, from the confusion matrix shown in Fig. 7, it is interesting that the *Submarine* with the smallest samples can be almost completely recognized while the *Cargo* with the largest samples would be predicted with

misunderstandings, e.g., *Bulker*, *Tanker*, and even background. In such a case, we may try to draw a conclusion that marine vessel detection is highly associated with feature saliency rather than data quantity, which is strongly in line with human-being visual recognition.

To intensively verify detection ability of occluded targets, within the entire testing dataset, 1173 occluded ships (OS) in 625 images are exclusively labeled with “occlusion=1”. In this context, testing dataset is divided into 6470 unoccluded and 625 occluded images, i.e., UI and OI. Accordingly, as summarized in Table V, the mAP metrics under different IoU thresholds are deployed to evaluate the detection capability for OS and unoccluded ships (US), respectively, so as to show that the AodeMar outperforms in both OS and US detection. Within UI dataset, comparing to the Top5 best-performing detectors shown in Table IV, the AodeMar preserves higher $mAP@[.5:.95]$, sub-optimal $mAP@.5$ and the largest number of correctly predicted targets, thereby demonstrating superiority in unoccluded scenarios. Conducting on OI dataset, the AodeMar outperforms with $\sim 1\%$ $mAP@.5$, and maximally recognizes 738 OS among 1173 samples in total. It can be evidently unfolded that, with higher precision and success rate, the AodeMar is uniformly able to detect both OS and US.

D. Real-World Validations

To thoroughly illustrate superiority of the proposed AodeMar in the real-world occlusion scenarios, visual detection

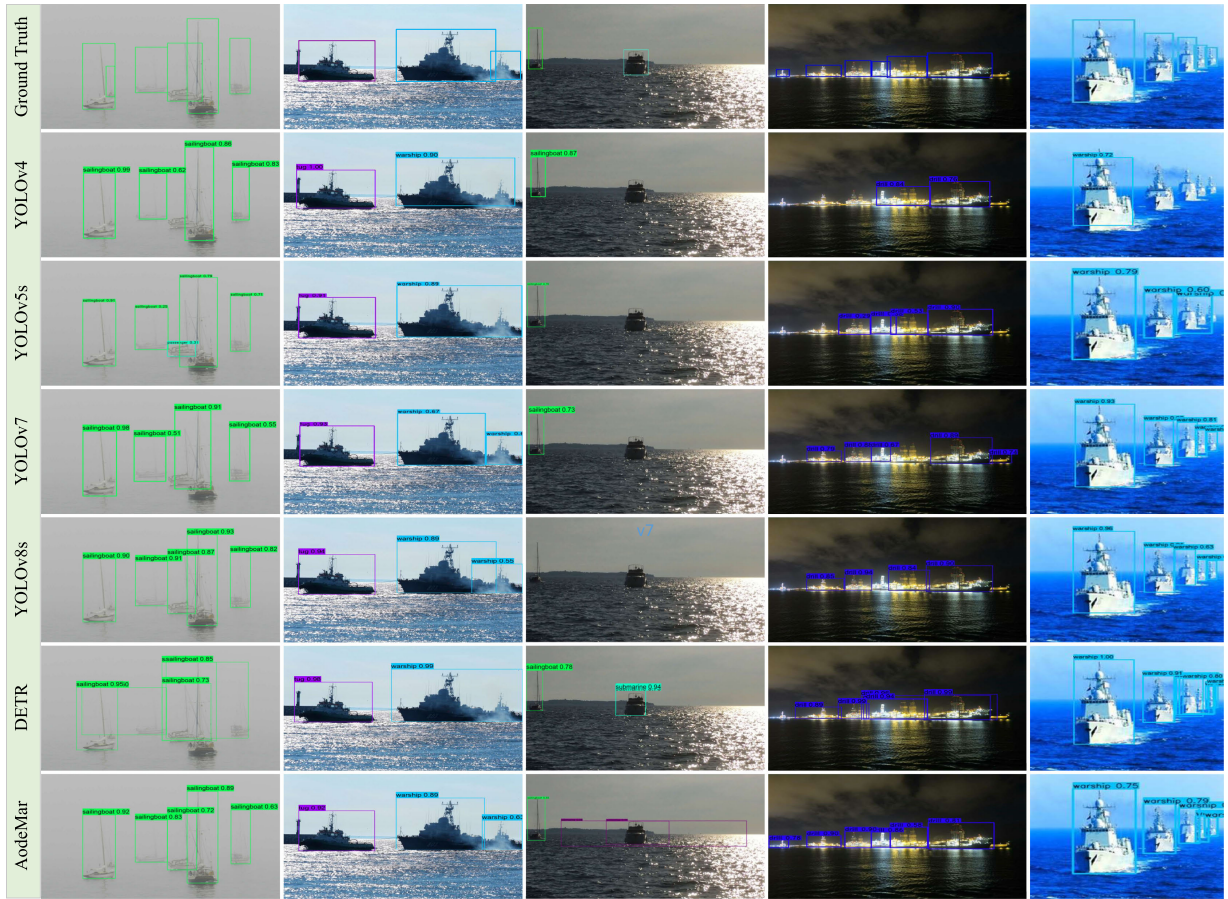


Fig. 9. Detection results in occlusion scenarios with different environmental disturbances, including 1) heavy sea fog, 2) bright light and wake streams, 3) outside the bright light area, 4) nighttime, and 5) color distortion.

results are shown in Fig. 8. The predicted information of each BBox in the image including coordinates, class and confidence score are employed to validate detection performance. Specifically, the following criteria are considered: 1) the closer the predicted box (or location) is to the object, the better the performance; 2) the predicted class that the object belongs to must be correct; 3) the higher the confidence score, i.e., the probability that object belongs to the predicted label, the better the classification performance.

It can be obviously seen that the AodeMar outperforms in 5 scenarios. As shown in the first column, the single target on the right is easy to be recognized, while 2 class-different and overlapping targets on the left are difficult. The AodeMar can accurately detect the small *Tug* that is connected to the *Tanker*, while counterparts fail to locate it. In this context, the superiority of the AodeMar in locating occluded ships is validated since the CA can learn long-range feature dependencies and enhance the association of local features. Additionally, for the second column, all targets have been located by the DETR, while the occluded *Containership* on the left is mistaken as *Tanker* and the one on the right is recognized as two objects. It should be highlighted that the AodeMar can correctly detect all targets with higher confidence scores. In berthing scenario shown in the third column, detectors are basically able

to detect ships on the right side that are close to the camera, but only the YOLOv5s and AodeMar are able to recognize smaller ones on the left side that are far away. For the fourth column, we can find that detectors perform well in recognizing *Sailingboat* with apparent sail-feature. In comparisons, YOLOv5s, YOLOv7 and YOLOv8s can recognize without generating redundant BBoxes, but lose more occluded ships. Comparing to the DETR generating massive redundant BBoxes, the AodeMar parsimoniously produces 2 proposals correctly detecting occluded ships. In the fifth scenario, there are 5 small and distant ships occluded by *Fishing*, for which all detectors perform well, except that the DETR generates an excessively redundant BBox. For the occluded small ships, the single one on the right can be correctly detected, as well as 2 targets at the middle are missed by all detectors, while there are two ships on the left side can only be accurately located by the AodeMar. Overall, the AodeMar shows comprehensive superiority in various real-world scenarios.

E. Robustness Performance Validations

To validate robustness of the AodeMar under environmental disturbances, 5 extreme sea-surface scenarios including heavy fog, ships in different bright light areas, nighttime, and color distortion are deployed to make comparisons.

The visual detection results are shown in Fig. 9, from which we can see that the AodeMar exhibits strong capabilities of avoiding lose detections. Concretely, in heavy fog scenario, the AodeMar and YOLOv8s achieve comparable results. In strong light reflection area, the AodeMar and YOLOv7 are good at positioning and classification, while the former achieves higher confidence scores. Note that, for those outside strong light reflection area, all models detect the ship on the left except for the YOLOv8s. For the most challenging one at the middle, only the AodeMar and DETR can recognize. In nighttime scene where visibility is severely restricted, it is evident that the AodeMar can detect more ships. Additionally, in scenario with color distortion and dense distributions, the AodeMar, YOLOv7 and YOLOv8s achieve comparable detection results, demonstrating their stability under such sophisticated conditions. In sum, the AodeMar is capable of maintaining robust detection performance under harsh marine environments.

V. CONCLUSION

In this paper, to facilitate occluded visual detectors for the MASS, an attention-aware occlusion detection scheme of marine vessels, termed AodeMar, has been developed. To that end, with the aid of the CA mechanism, position enhancement module has been devised while multi-scale feature semantics correlation module has been created by virtue of the STR. As a consequence, accurate localization and precise classification of occluded ships can be achieved, simultaneously, thereby fertilizing visual detection ability of the MASS under occluded circumstances. Comprehensive experiments and comparisons demonstrate that the proposed AodeMar framework can efficiently outperform the typical approaches in terms of precision and completeness. In future work, drastic occlusions, e.g., visible parts of a occluded object are disconnected, would be exclusively investigated.

ACKNOWLEDGMENT

The authors would like to thank the Editor-in-Chief, an Associate Editor, and anonymous reviewers for their invaluable suggestions and comments.

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Ning Wang (Senior Member, IEEE) received the B.Eng. degree in marine engineering and the Ph.D. degree in control theory and engineering from Dalian Maritime University, Dalian, China, in 2004 and 2009, respectively.

From September 2008 to September 2009, he was financially supported by China Scholarship Council to work as a joint-training Ph.D. Student with Nanyang Technological University (NTU), Singapore. From August 2014 to August 2015, he worked as a Visiting Scholar with The University of Texas at San Antonio. He has published more than 200 SCI-indexed journal articles. His research interests include self-learning autonomy, unmanned (marine) vehicles, and autonomous systems. He is currently a fellow of the Institution of Engineering and Technology (IET), the Institute of Marine Engineering, Science & Technology (IMarEST), and The International Society for Condition Monitoring (ISCM). He was honored with the National High-Level Talents Program for Young Top-Notch Talents. In view of his significant research with NTU, he received the Excellent Government-Funded Scholars and Students Award in 2009. He was recognized as the Highly Cited Researcher by Clarivate Analytics and the Highly Cited Chinese Researcher in Naval Architecture and Ocean Engineering by Elsevier. He has been the Leading Guest Editor of *International Journal of Robust and Nonlinear Control*, *Neurocomputing*, *Control Engineering Practice*, *IFAC Mechatronics*, *IET Control Theory & Applications*, and *International Journal of Fuzzy Systems*. He serves as an Associate Editor for IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE SIGNAL PROCESSING LETTERS, IEEE SYSTEMS JOURNAL, *Ocean Engineering*, *ISA Transactions*, *International Journal of Fuzzy Systems*, *Transactions of the Institute of Measurement and Control*, *IET Electronics Letters*, and *Journal of Electrical Engineering & Technology*.

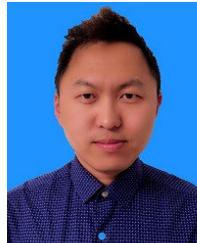


Yuanyuan Wang received the M.S. degree in applied mathematics from Dalian Maritime University (DMU), Dalian, China, in 2017, where she is currently pursuing the Ph.D. degree in marine electric engineering.

Her current research interests include deep learning, object detection, and image processing.



Yuan Feng received the B.Eng. degree in heating, ventilation, and air conditioning from Xi'an University of Architecture and Technology, Xi'an, China, in 2020. He is currently pursuing the M.Eng. degree in marine engineering with Dalian Maritime University. His current research interests include deep learning, object detection, and image processing.



Yi Wei received the master's and Ph.D. degrees in marine engineering from Dalian Maritime University, Dalian, China, in 2007 and 2012, respectively. From January 2015 to June 2015, he worked as a Visiting Scholar with Washington University in St. Louis. From June 2015 to January 2016, he worked as a Visiting Scholar with the University of California at Berkeley. His research interests include machine learning, renewable energy, fault diagnosis of marine electromechanical equipment, and new material research machine learning. He has

authored two books, he currently serves as the Guest Editor for *Transactions of Csice*.